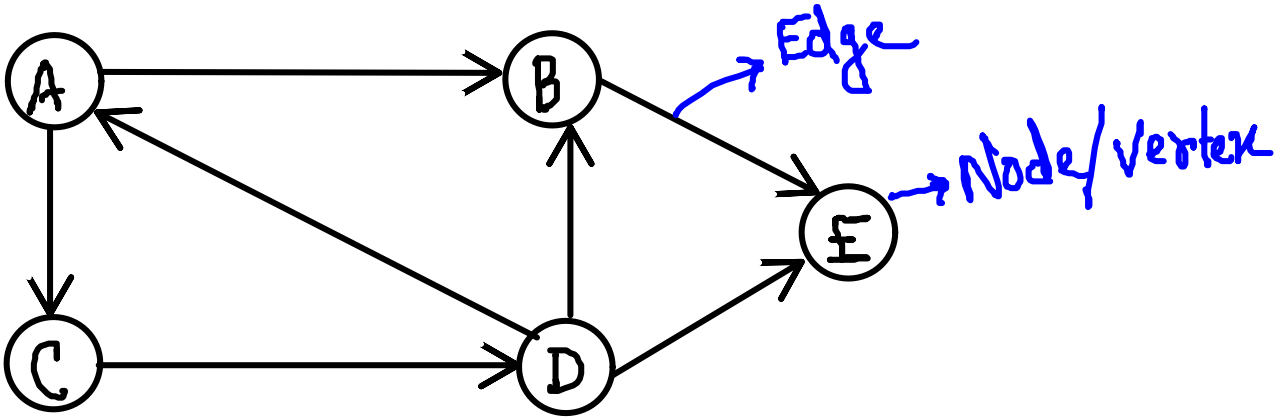
Graph



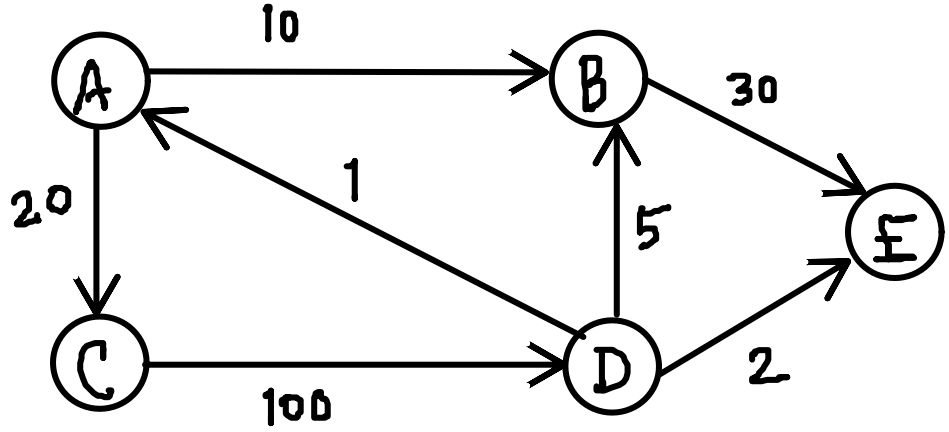
- Unweighted
- Weighted
- Undirected
- Directed

Unweighted Adjacency Matrix:

	A	B	C	D	E
A	0	1	1	0	0
B	0	0	0	0	1
C	0	0	0	1	0
D	1	1	0	0	1
E	0	0	0	0	0

1 → Connection

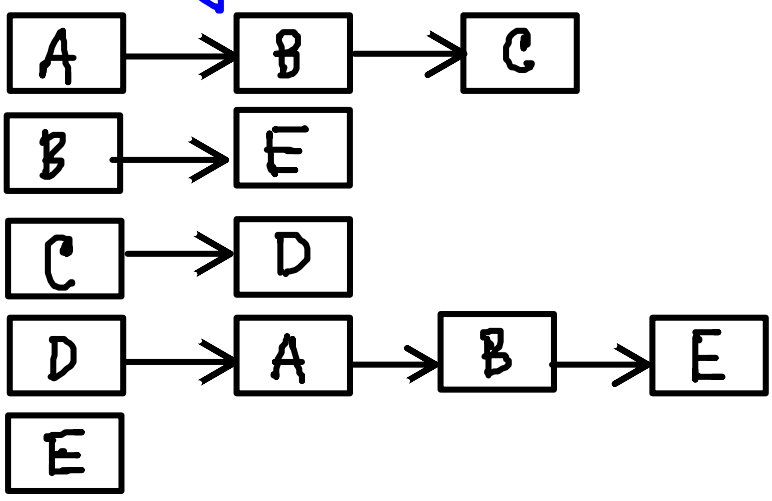
0 → No connection



Weighted Adjacency Matrix:

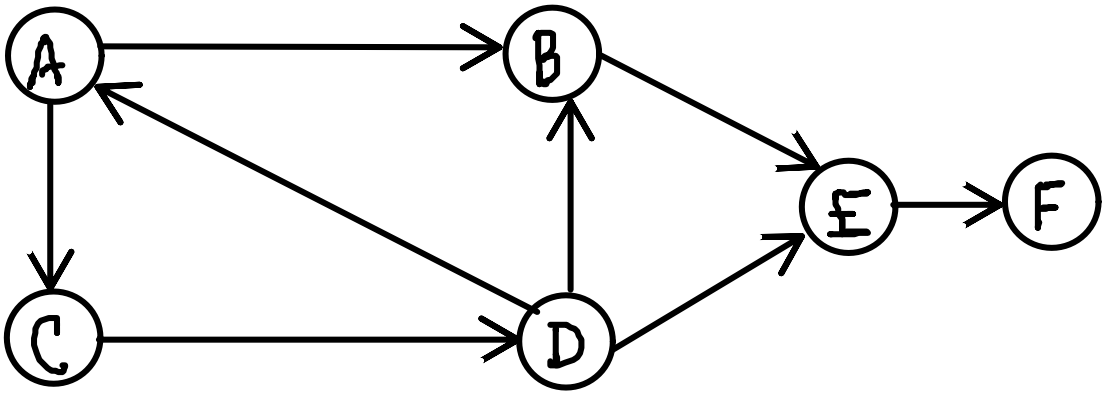
	A	B	C	D	E
A	0	10	20	0	0
B	0	0	0	0	30
C	0	0	0	100	0
D	1	5	0	0	2
E	0	0	0	0	0

Adjacency List :



Graph Traversal

Breadth First Search (BFS)



	A	B	C	D	E	F
Visited =	1	1	1	1	1	1

	A	B	C	D	E	F
Parent =	null	A	A	C	B	E

	A	B	C	D	E	F
Distance =	0	0+1	0+1	1+1	1+1	2+1

By default value 0/null

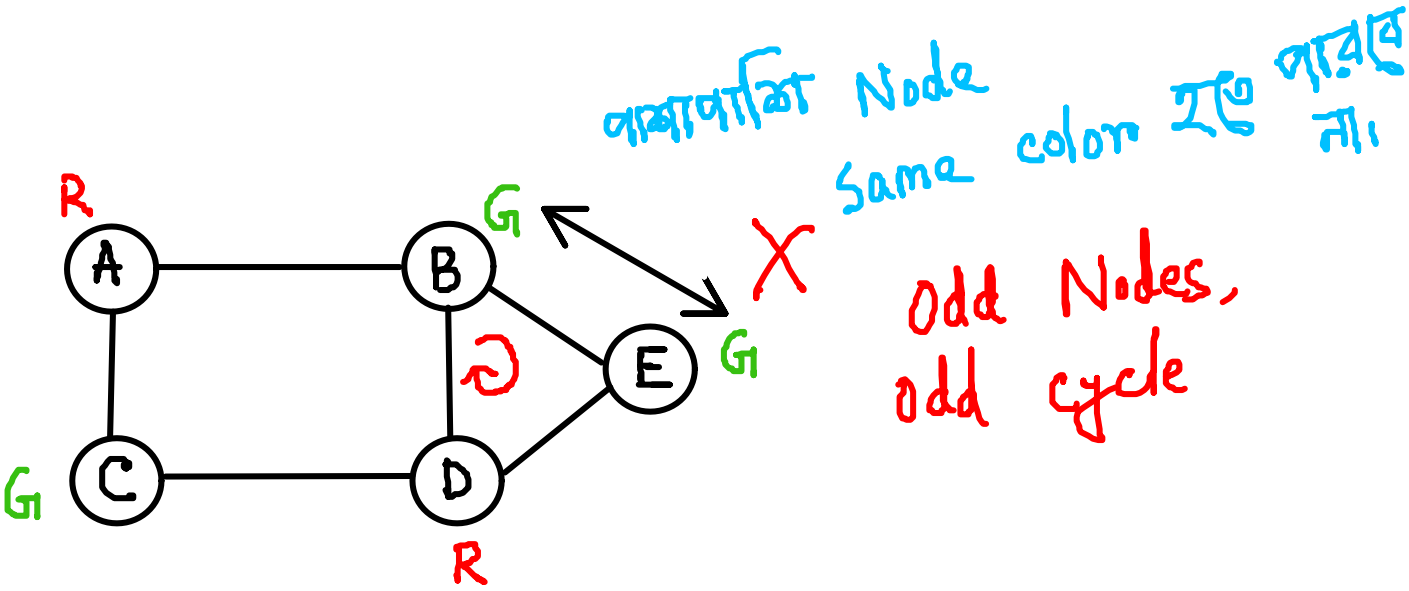
Queue : ~~A~~ ~~B~~ ~~C~~ ~~E~~ ~~D~~ ~~F~~ ~~F~~

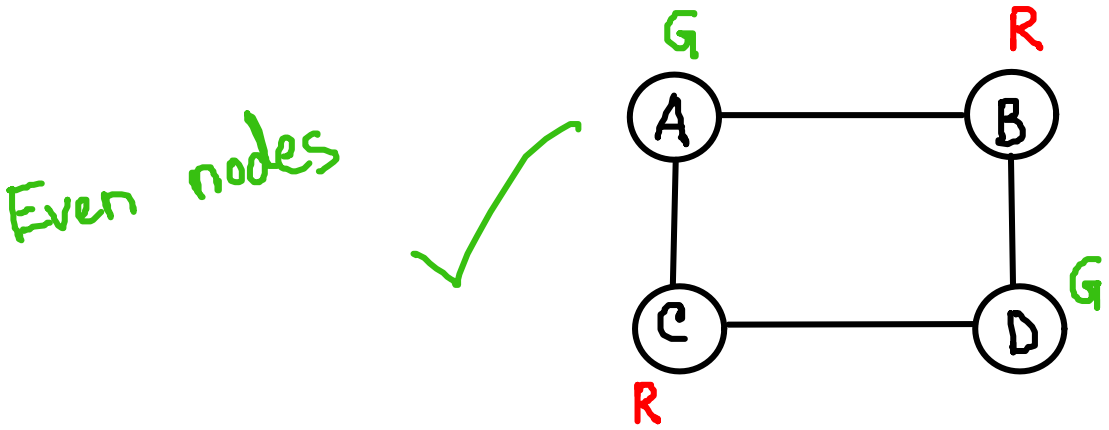
Shortest Path:

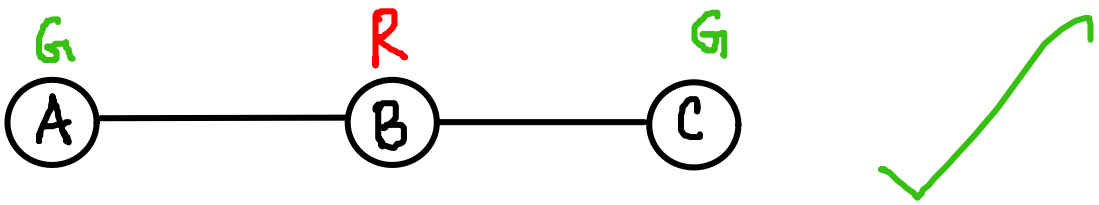
Node F : A → B → E → F

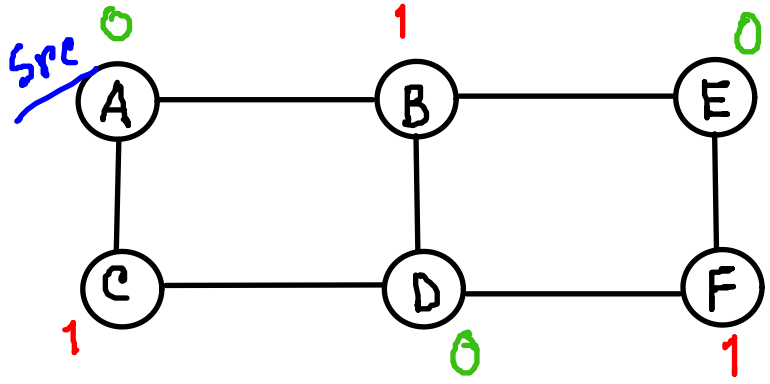
Node D : A → C → D

Bipartite Graph / Bicolorable Graph









By default value 0/null

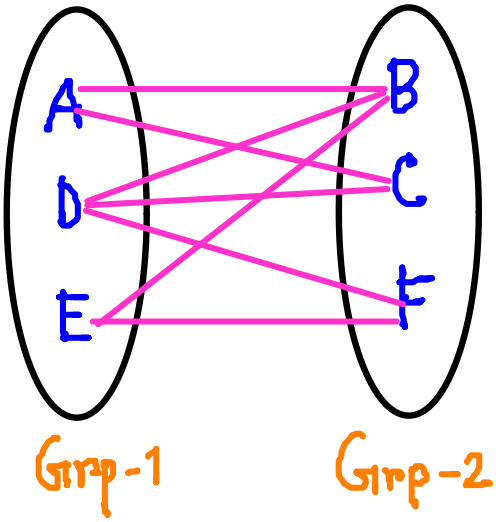
R → 0

G → 1

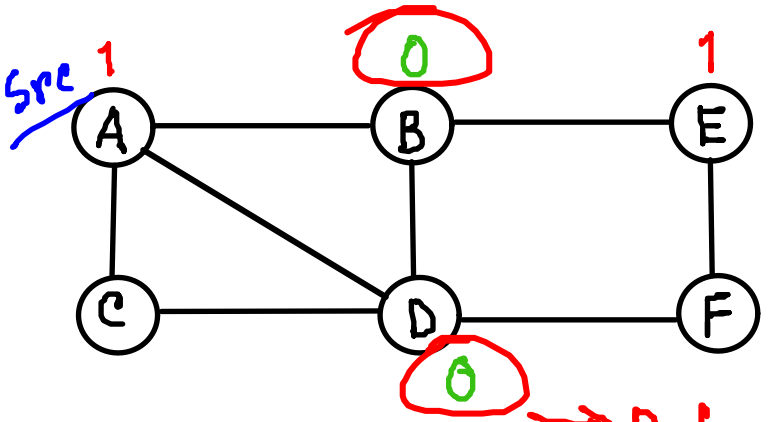
	A	B	C	D	E	F
Color =	0	1	1	0	0	1

	A	B	C	D	E	F
Parent =	null	A	A	B	B	D

	A	B	C	D	E	F
Distance =	0	0+1	0+1	1+1	1+1	2+1



Queue = ~~A~~ ~~B~~ ~~C~~ ~~D~~ ~~E~~ ~~F~~



→ Return False